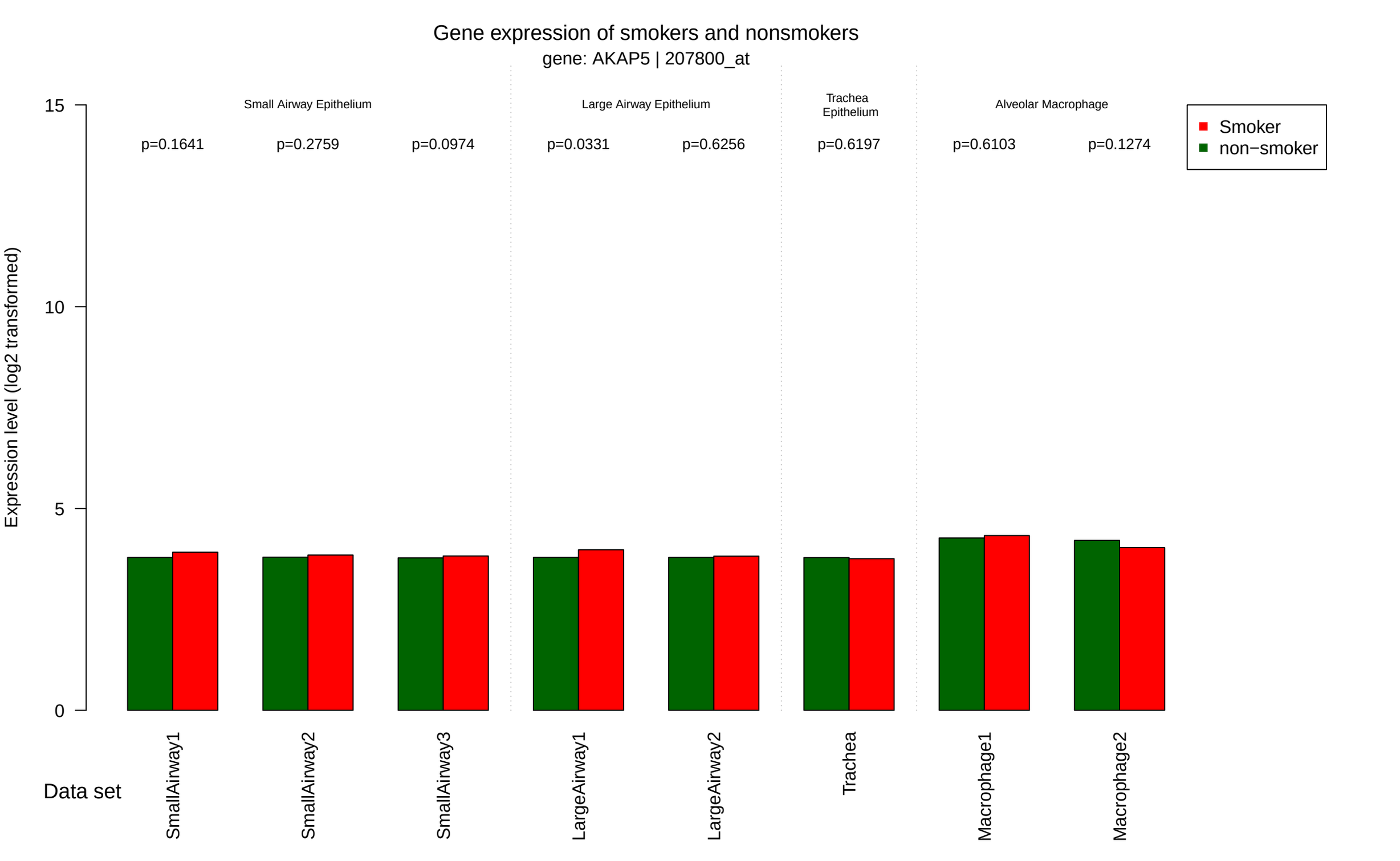
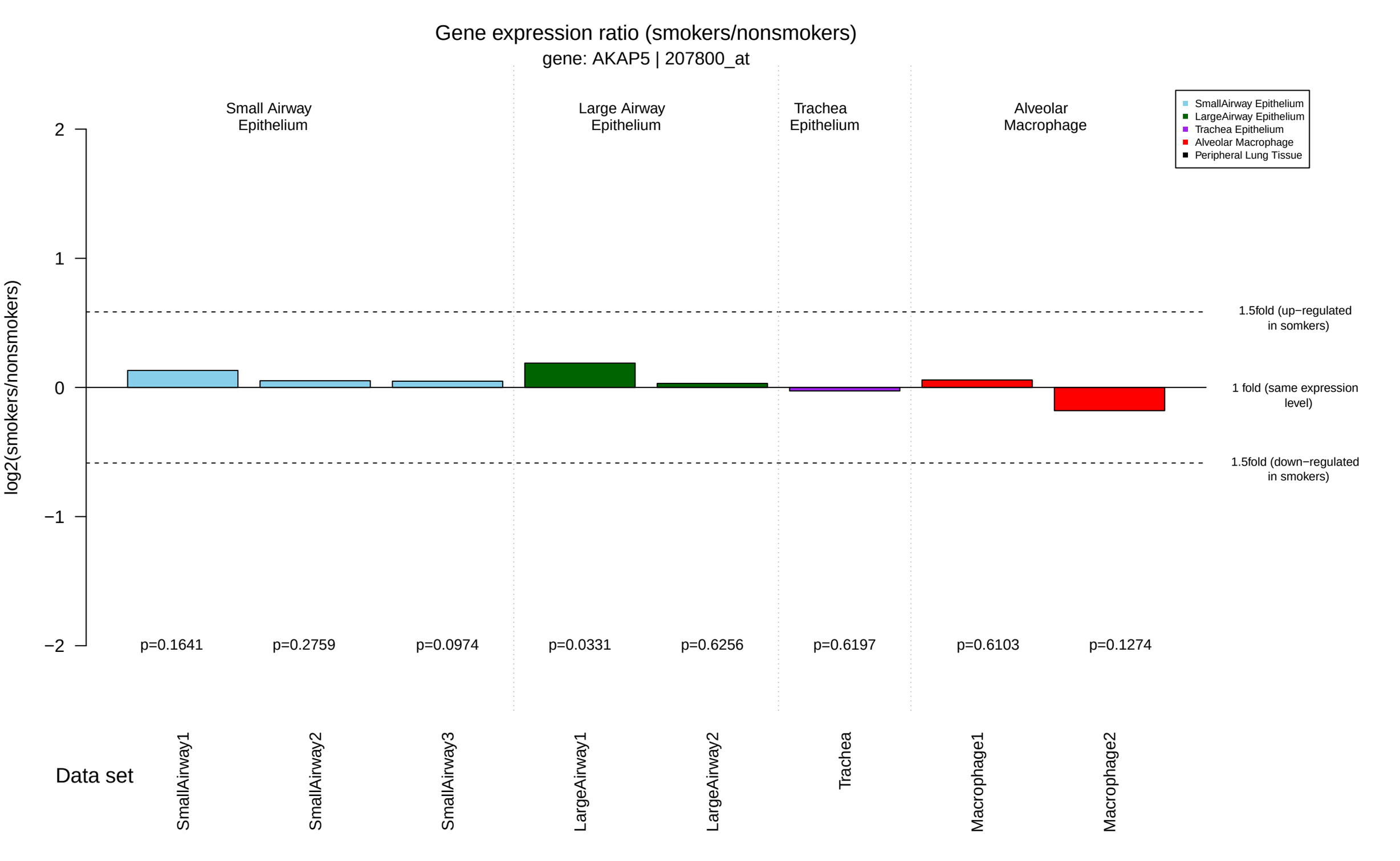
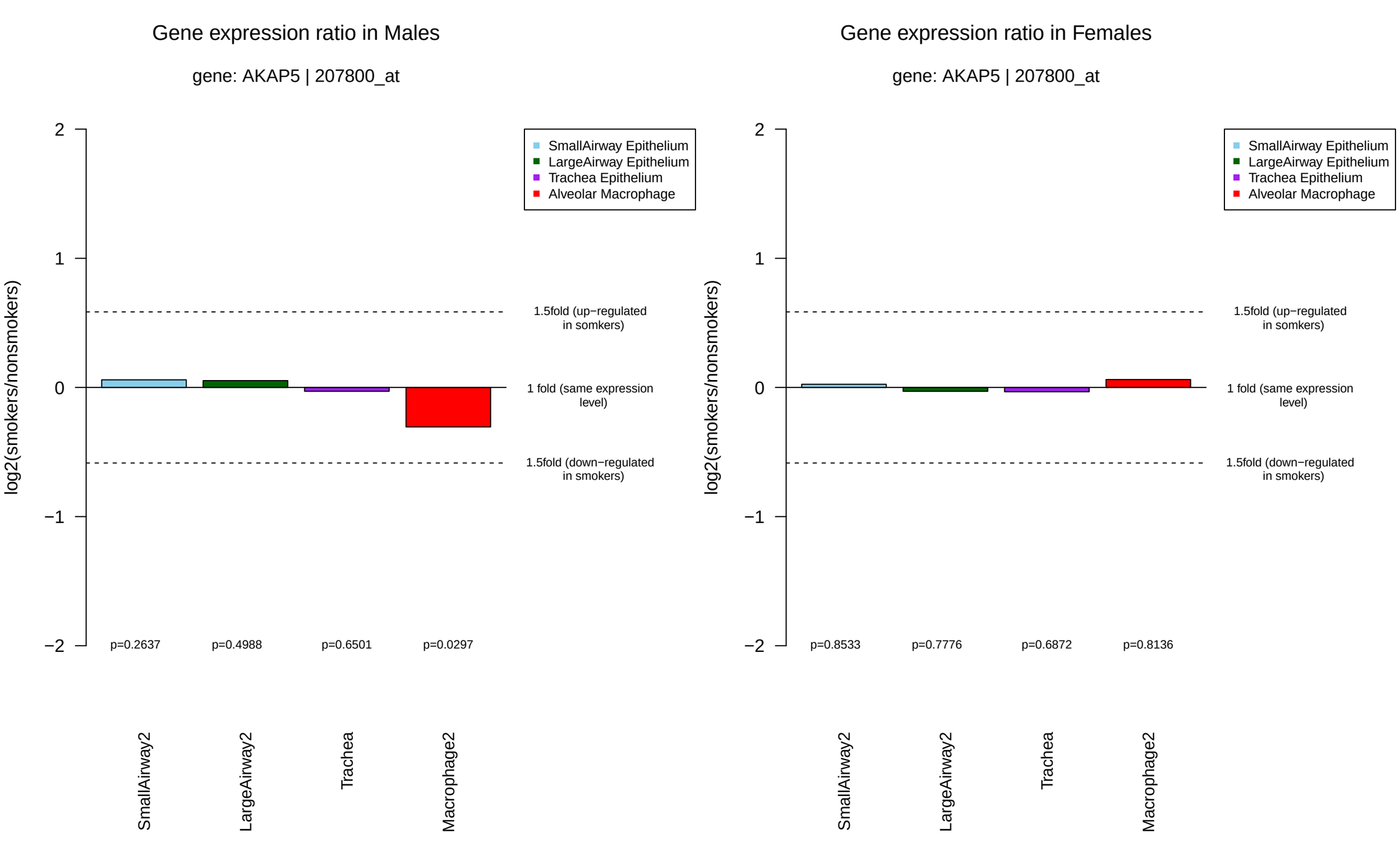
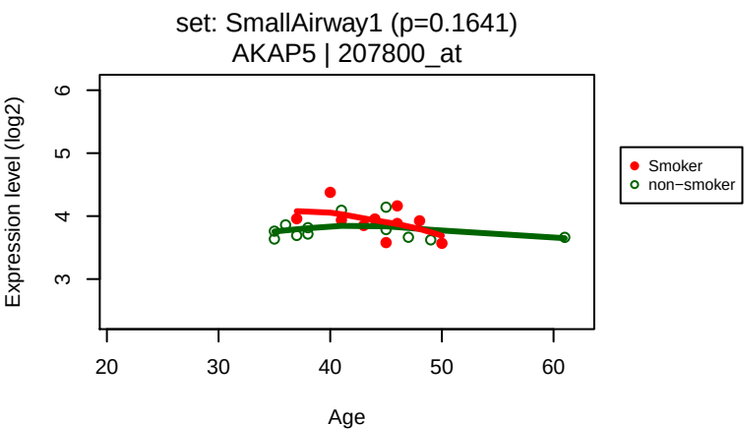
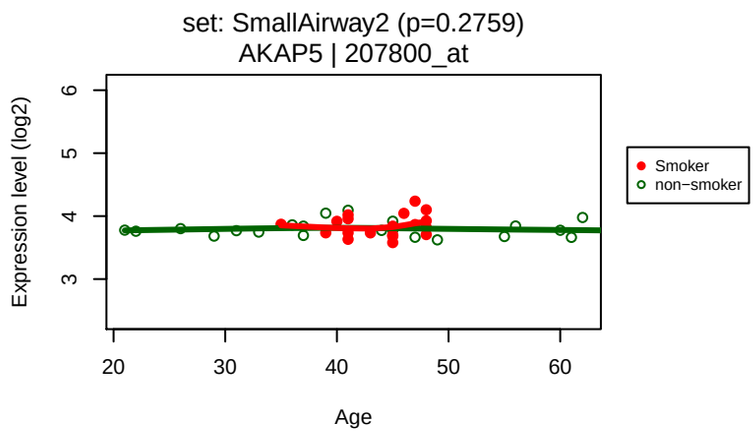
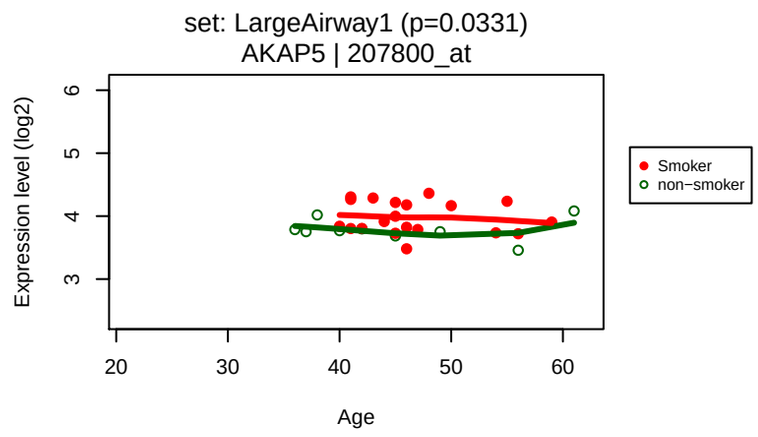

Gene expression of smokers and nonsmokers
gene: AKAP5 | 207800_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1641
p=0.2759
p=0.0974
p=0.0331
p=0.6256
p=0.6197
p=0.6103
p=0.1274
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

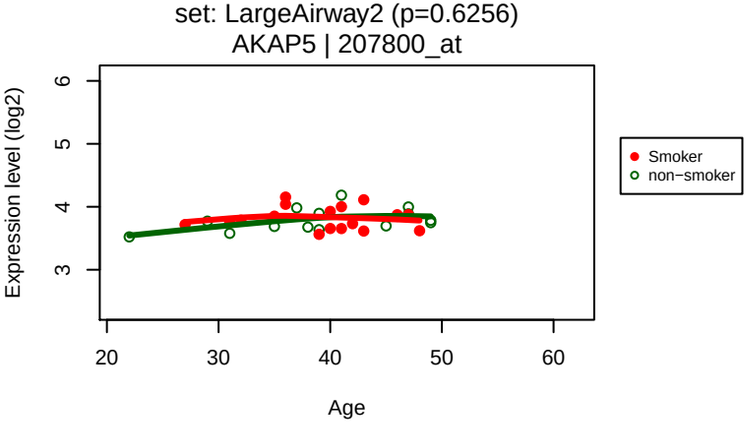

Gene expression ratio (smokers/nonsmokers)
gene: AKAP5 | 207800_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1641
p=0.2759
p=0.0974
p=0.0331
p=0.6256
p=0.6197
p=0.6103
p=0.1274
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

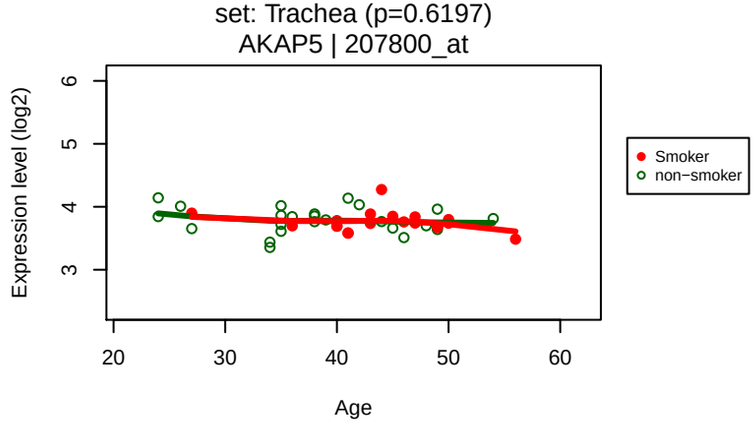

Gene expression ratio in Males
gene: AKAP5 | 207800_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2637
p=0.4988
p=0.6501
p=0.0297
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AKAP5 | 207800_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8533
p=0.7776
p=0.6872
p=0.8136
SmallAirway2
LargeAirway2
Trachea
Macrophage2

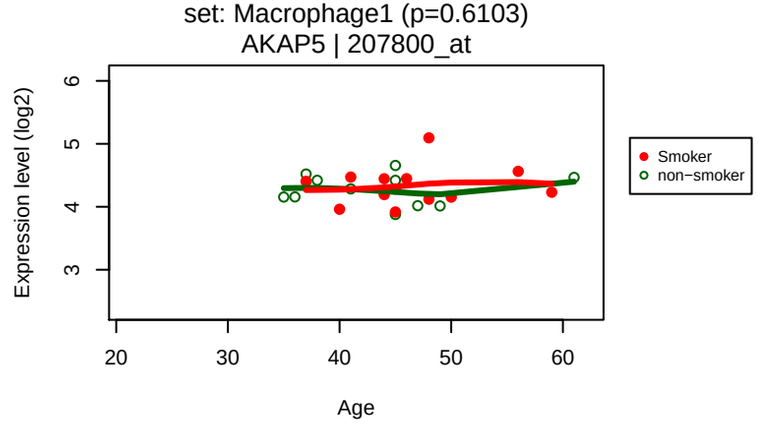

set: SmallAirway1 (p=0.1641)
AKAP5 | 207800_at
Expression level (log2)
Age
Smoker
non−smoker

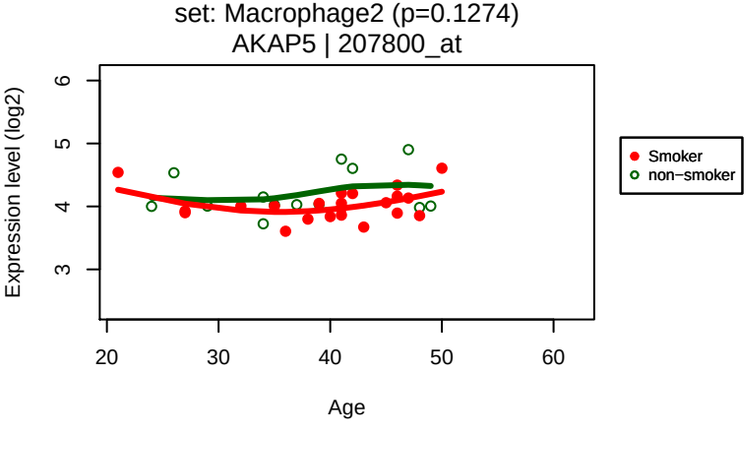

set: SmallAirway2 (p=0.2759)
AKAP5 | 207800_at
Expression level (log2)
Age
Smoker
non−smoker

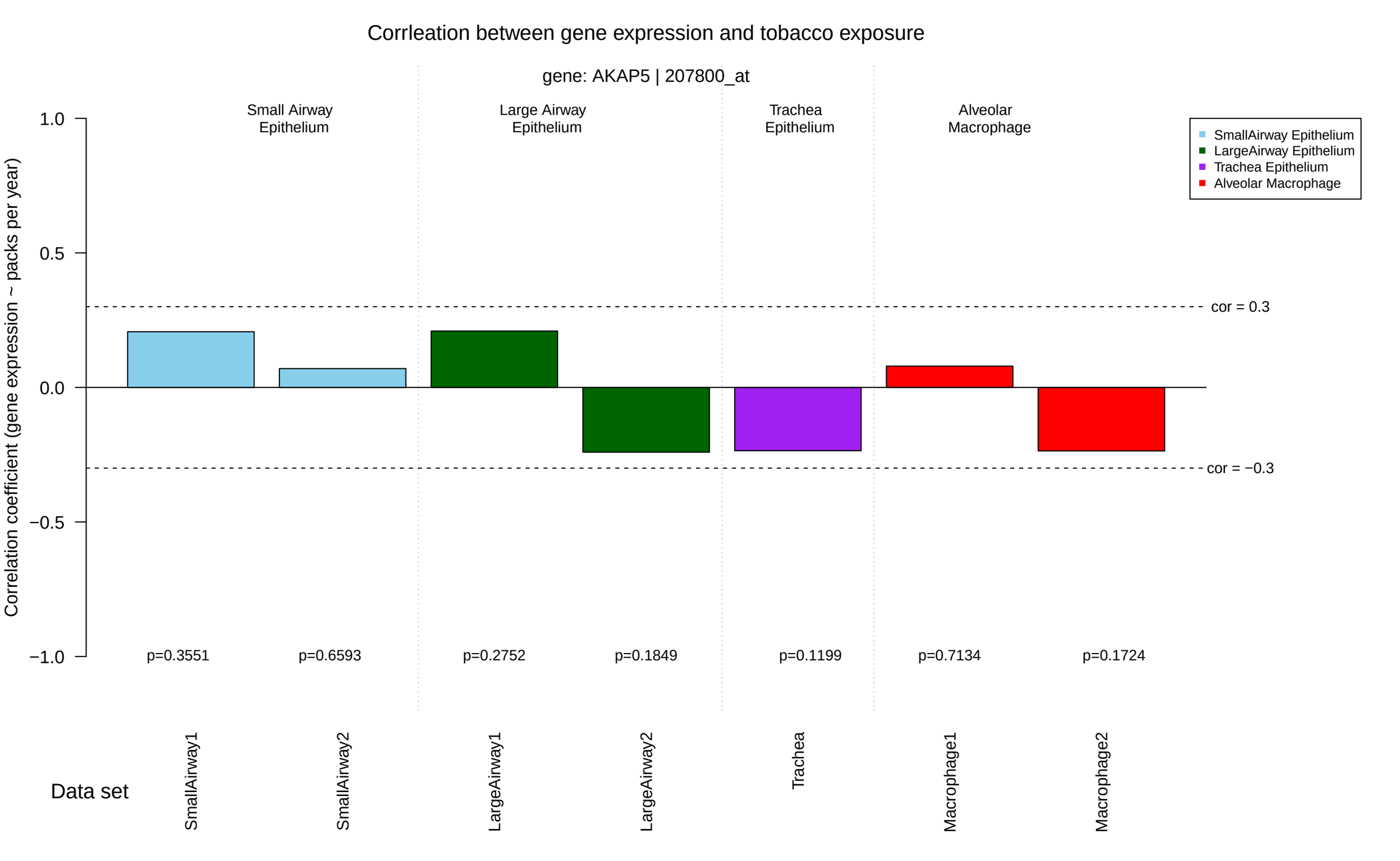

set: LargeAirway1 (p=0.0331)
AKAP5 | 207800_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6256)
AKAP5 | 207800_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6197)
AKAP5 | 207800_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6103)
AKAP5 | 207800_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1274)
AKAP5 | 207800_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AKAP5 | 207800_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3551
p=0.6593
p=0.2752
p=0.1849
p=0.1199
p=0.7134
p=0.1724
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2